HAMED SAHEBI

Software Engineer | System Integration & Data Architecture

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PROFILE

Software engineer with 5 years designing enterprise data systems for energy operations and drilling workflows. Founded on **M.Sc. Well Engineering**; expertise in **Python**, **data architecture**, and **system design** for production-critical operations. Proven ability to translate energy sector requirements into scalable solutions, align stakeholders on system architecture, and secure funding through compelling business cases. Track record of securing 5M and 500K NOK in research funding for operational technology.

RELATED SKILLS

System Design, **Data Architecture**, **REST API Design**, **PostgreSQL**, **MongoDB**, **Docker**, **Python**, **Stakeholder Alignment**, **System Integration**, **InfluxDB**

EXPERIENCE

Software Engineer

University of Stavanger

2020 – Present Stavanger

AI Integration for Energy Operations (2023 – Present)
Leading solution design for AI-driven systems supporting critical energy operations

- Held end-to-end responsibility for AI solution architecture; translated business and technical requirements into architectural blueprints and system specifications for production deployment.
- Collaborated with stakeholders, domain experts, and implementation teams to guide specification, solution design, and ensure alignment with business objectives and operational requirements.
- Assessed AI feasibility pragmatically across energy sector use cases; communicated capabilities, limitations, and integration challenges to technical and business stakeholders.
- Created architectural documentation and design guidelines; ensured implementation adherence to system architecture standards and enterprise principles.

Drilling Data Platform & System Architecture (2020 - 2021)
Led solution design for data-intensive systems supporting critical drilling operations

- Held holistic responsibility for end-to-end architecture and design; ensured alignment between solution design and operational business objectives throughout implementation.
- Proactively led specification and solutioning of drilling operational workflows; collaborated with operations teams, subject matter experts, and system stakeholders on requirements translation and design validation.
- Evaluated and translated operational requirements into architectural blueprints; designed **data models** and **system integration** strategies supporting high-frequency critical drilling telemetry.
- Created comprehensive architectural documentation; guided implementation teams on system design standards and production reliability approaches.
- Secured 5M NOK (2024–2026) and 500K NOK (2022–2023) research funding through solution design excellence and business case development for operational technology innovation.

Software Developer

Valide As

June 2019 – December 2019 Stavanger

Data processing and signal analysis for energy system monitoring

- Developed **Python** applications for energy equipment data analysis and automated signal processing.
- Implemented feature extraction and filtering techniques using **NumPy** and **pandas** for operational pattern recognition.
- Designed databases for structured operational data storage and real-time retrieval across energy systems.

ML Engineer Intern

ProWellPlan As

June 2018 – December 2018 Bergen

Predictive modeling for drilling and well planning operations

- Developed **machine learning** models for well trajectory prediction and operational forecasting using time-series analysis.
- Engineered features from drilling datasets; validated model performance on production well data and planning workflows.

TECHNICAL SKILLS

Languages: Python, JavaScript
Backend & APIs: FastAPI, Flask, REST APIs, system integration
Infrastructure & DevOps: Docker, Kubernetes, Linux, Git, GitHub Actions, CI/CD pipelines
Databases: PostgreSQL, InfluxDB, MongoDB
AI Tools: LLMs, Autonomous Agents, LangChain
Specializations: Energy data systems, drilling operations technology, backend systems, production reliability, operational data architecture

EDUCATION

M.Sc. Well Engineering

University of Stavanger

Aug 2017 – Jun 2020

Drilling engineering, well design, operational technologies **GPA: A**

Computer Science

University of Oslo

Jan 2021 – Jun 2021

High-performance computing and parallel programming using C++ **GPA: B**

CERTIFICATES

- **Docker & Kubernetes** (22h) – Containerization, orchestration, and infrastructure deployment
- **LangChain & LangGraph** (18.5h) – AI agents and autonomous workflow automation
- **Python for Data Science** (21.5h) – Data processing and system validation
- **Node.js & REST APIs** (15h) – Backend architecture and service integration
- **Understanding TypeScript** (15h) – Type-safe development and full-stack integration

LANGUAGES

English – Fluent | Norwegian – B1

REFERENCES

- **Øystein Arild** – Head of Department, Energy & Petroleum Engineering, University of Stavanger
- **Dan Sui** – Professor, University of Stavanger